

**Cambridge IGCSE™**CANDIDATE
NAMECENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

BIOLOGY**0610/62**

Paper 6 Alternative to Practical

February/March 2025**1 hour**

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Any blank pages are indicated.



- 1 Benedict's solution is used to test for reducing sugars. Glucose is a reducing sugar. A student estimated the concentration of glucose in a solution.

The student used this method:

Step 1 Label six test-tubes **0.0%**, **0.5%**, **1.0%**, **1.5%**, **2.0%** and **U**.

- (a) (i) Complete Table 1.1 by writing in the volumes of 2% glucose solution and distilled water needed to make 4 cm³ of a 1.5% glucose solution.

Table 1.1

percentage concentration of glucose	volume of 2% glucose solution / cm ³	volume of distilled water / cm ³
0.0	0	4
0.5	1	3
1.0	2	2
1.5		
2.0	4	0

[1]

- Step 2 Use a syringe to put the volumes of 2% glucose solution shown in Table 1.1 into the test-tubes labelled **0.5%**, **1.0%**, **1.5%** and **2.0%**.
- Step 3 Use the same syringe to put 4 cm³ of the unknown glucose solution **U** into the test-tube labelled **U**.
- Step 4 Use a clean syringe to put the volumes of distilled water shown in Table 1.1 into the test-tubes labelled **0.0%**, **0.5%**, **1.0%**, and **1.5%**.
- Step 5 Use a clean syringe to put 4 cm³ of Benedict's solution into each of the test-tubes labelled **0.0%**, **0.5%**, **1.0%**, **1.5%**, **2.0%** and **U**.
- Step 6 Put the test-tubes into a hot water-bath at 80 °C and start the stop-clock.
- Step 7 Wait for 5 minutes. Remove all the test-tubes from the water-bath.
- Step 8 Record the colour of the liquid in the test-tubes labelled **0.0%**, **0.5%**, **1.0%**, **1.5%**, **2.0%** and **U**.



Fig. 1.1 shows the student's results.

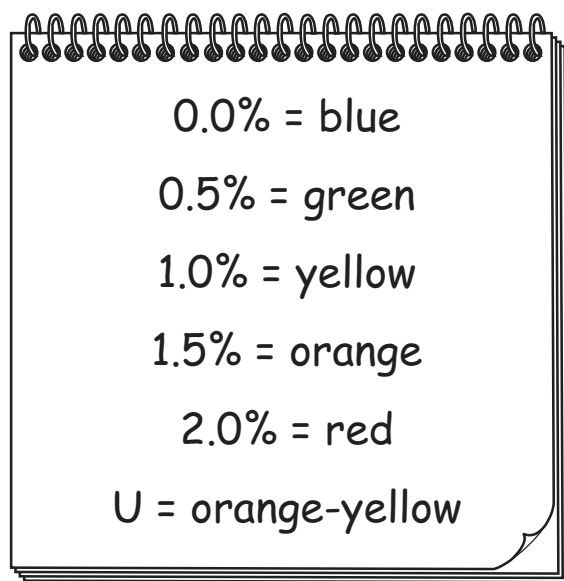


Fig. 1.1

- (ii) Prepare a table and record the student's results.

[3]

- (iii) Use Fig. 1.1 to estimate the percentage concentration of glucose in the unknown glucose solution **U**.

concentration of glucose % [1]





(iv) State **two** variables that were kept constant in this investigation.

1

2 [2]

(v) Identify the possible source of error present in steps 2 and 3.

State the effect of this error on the results of the investigation.

error

.....

effect on the results

.....

..... [2]

(b) Eggs contain a protein called albumen. Albumen will turn from cloudy to clear when it is digested by pepsin, a protease enzyme.

Plan an investigation to determine the effect of pH on the digestion of albumen by protease.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]





(c) Describe how to do the emulsion test to show that fat is present in a food sample.

.....

.....

.....

..... [2]

[Total: 17]



- 2 (a) Fig. 2.1 is a photograph of a leaf from an oak tree, *Quercus* sp.

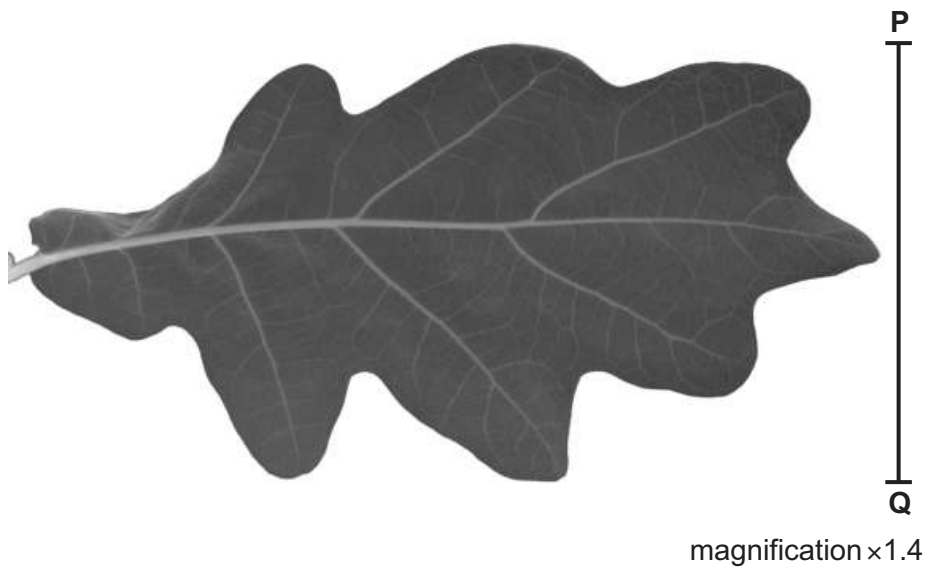


Fig. 2.1

- (i) Draw a large diagram of the whole oak leaf shown in Fig. 2.1.

[4]





(ii) Line **PQ** on Fig. 2.1 represents the width of the oak leaf.

Measure the length of line **PQ** on Fig. 2.1.

length of **PQ** mm

Calculate the actual width of the oak leaf using the formula and your measurement.

$$\text{magnification} = \frac{\text{length of line } \mathbf{PQ} \text{ in Fig. 2.1}}{\text{actual width of the oak leaf}}$$

Give your answer to **two** significant figures.

Space for working.

..... mm
[3]





(iii) Fig. 2.2 shows photographs of the oak leaf and a hollyhock (*Alcea rosea*) leaf.

The magnification of both photographs is **not** the same.

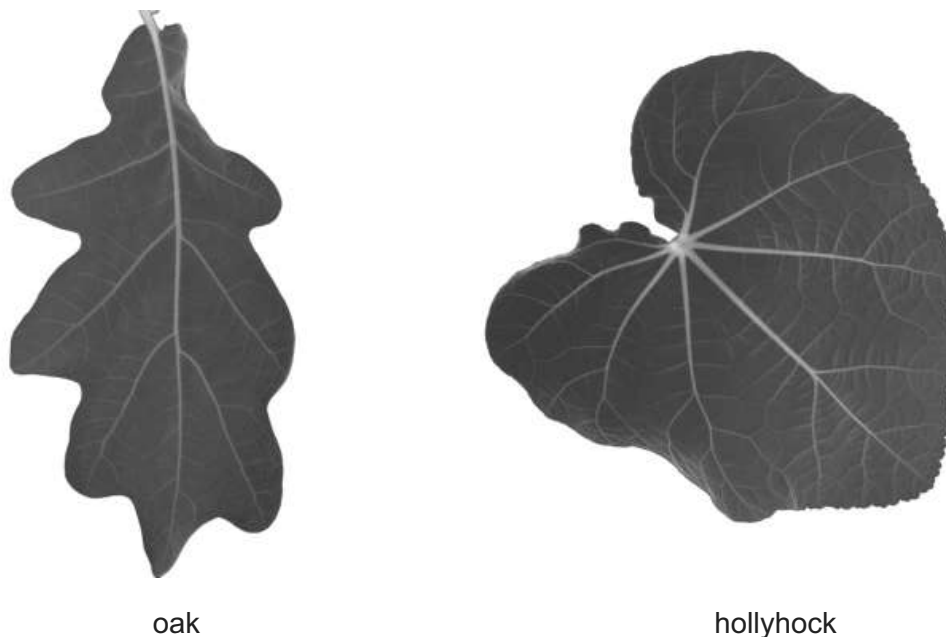


Fig. 2.2

Table 2.1 gives one visible difference between the oak leaf and the hollyhock leaf shown in Fig. 2.2.

Complete Table 2.1 to give **two other** visible differences between the leaves shown in Fig. 2.2.

Do **not** include references to size in your answer.

Table 2.1

difference	oak leaf	hollyhock leaf
1	The oak leaf is longer than it is wide.	The length and width of the hollyhock leaf are similar.
2		
3		

[2]



(b) Students investigated the effect of light intensity on the surface area of leaves of soybean plants.

- 100 soybean seeds were planted in pots and put into the shade (low light).
- 100 soybean seeds were planted in pots and put into full sun.
- The soybean seeds were allowed to germinate and grow for 30 days.
- After 30 days, three of the oldest leaves and three of the youngest leaves were removed from each plant.
- The surface area of each of the removed leaves was measured.

(i) State the dependent variable in this investigation.

..... [1]

(ii) State why the students used a large number of soybean plants.

.....
 [1]

(iii) The students estimated the surface area of each leaf using graph paper as shown in Fig. 2.3.

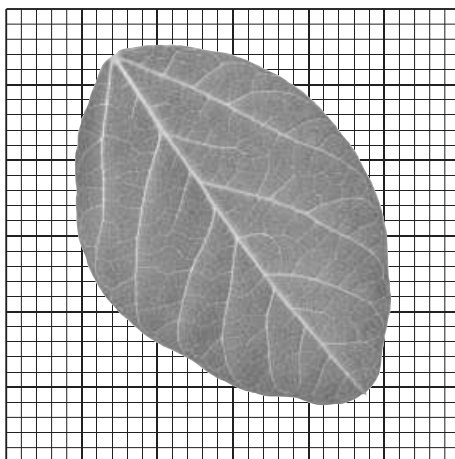


Fig. 2.3

Suggest how the students used the graph paper to measure the surface area of each leaf.

.....

 [2]



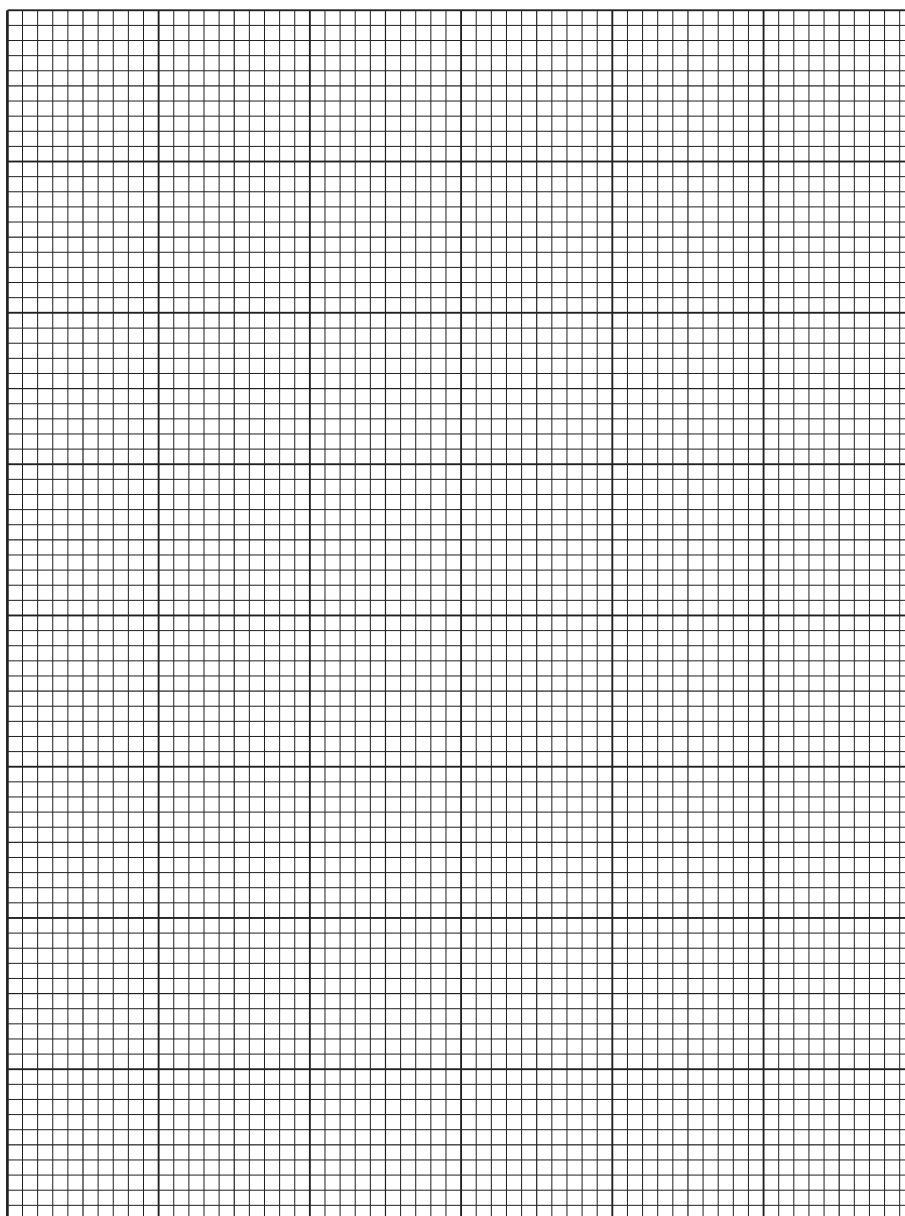


(c) The results of the investigation are shown in Table 2.2.

Table 2.2

lighting condition	age of leaves	mean surface area of leaves/cm ²
full sun	young	52
shade	young	45
full sun	old	78
shade	old	58

- (i) On the grid, using the data in Table 2.2, plot a bar chart to show the results of the investigation.



[4]



- (ii) The students calculated that there was a 15.6% increase in the size of the young leaves when they were grown in full sun compared to in the shade.

Calculate the percentage increase in the surface area of the old leaves that had been grown in full sun conditions compared to the old leaves that had been grown in the shade.

Give your answer to **one** decimal place.

..... % [3]

- (iii) State **two** conclusions from this investigation.

1

.....

2

.....

[2]

- (d) Hydrogencarbonate indicator is used to test for the presence of carbon dioxide.

An aquatic plant was placed in red hydrogencarbonate indicator and put under bright light. The plant takes in carbon dioxide as it photosynthesises.

State the final colour of the hydrogencarbonate indicator.

..... [1]

[Total: 23]





Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

